

A Mechanism-Informed Predictive Model for Mean and Sigma of 3D NAND Threshold-Voltage Distributions at Extreme Temperatures

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Abstract—Emerging aerospace and other mission-critical applications require charge-trap 3D NAND flash to operate reliably over wide and extreme temperature ranges, where the cross-temperature evolution of threshold-voltage (V_{TH}) distributions is strongly nonlinear and trajectory-dependent. In this work, we propose a mechanism-informed compact model that jointly predicts the mean and sigma of 3D NAND V_{TH} distributions by decomposing trap, barrier, and mobility contributions. The model is validated across programmed states under high-temperature program/low-temperature read and low-temperature program/high-temperature read trajectories over -55°C to 125°C . It achieves accurate full-window fitting for both descriptors, while restricted-window fitting and held-out-region prediction demonstrate robust extrapolation into extreme-temperature regimes. Ablation analysis shows that the barrier term contributes most to prediction accuracy, followed by the trap term, while the mobility term provides a smaller but visible contribution. Compared with empirical baselines, the proposed model reduces the average prediction RMSE by 66.9% in extreme-temperature and nonlinear regions, providing a compact framework for distribution-level prediction and read-threshold optimization in 3D NAND.

Index Terms—3D NAND flash, cross-temperature modeling, extreme temperature operation, mechanism-informed predictive model.

I. INTRODUCTION

THREE-DIMENSIONAL (3D) NAND flash memory has become the dominant high-density storage technology [1]. The mean and sigma of the threshold-voltage (V_{TH}) distribution jointly determine state overlap, guard-band margin, and read robustness [2], [3], [4], [5], [6]. Under cross-temperature operation, charge trapping, grain-boundary transport, and readout-condition changes drive nonlinear, state-

and trajectory-dependent V_{TH} -distribution evolution [2], [5], [7], [8], [9], especially toward extreme temperatures.

Although temperature effects in 3D NAND have been widely investigated [5], [6], [7], [8], [9], [10], existing models are mainly empirical, focus on mean shift, or are limited to narrow temperature windows [3], [5], [6], [7]. Therefore, a compact model that jointly predicts mean and sigma over a wide temperature range with physical interpretability is still needed. To improve commercial-grade NAND wide-temperature robustness via product-specific read-reference calibration instead of costly device/process hardening, we propose a physics-constrained predictive model for cross-temperature 3D NAND V_{TH} distributions evolution. The model jointly predicts descriptor-level mean and sigma under high-temperature program/low-temperature read (HPLR) and low-temperature program/high-temperature read (LPHR) trajectories. Validated on advanced 232-layer 3D NAND flash, the trap-barrier-mobility formulation agrees with full-window data, while restricted-window prediction and empirical-model comparison verify its nonlinear extreme-temperature extrapolation capability.

II. DERIVING THE PHYSICS-CONSTRAINED PREDICTIVE MODEL

To describe cross-temperature V_{TH} evolution in a unified mechanism-informed framework, the mean and sigma are decomposed into trap, barrier, and mobility contributions:

$$V_{TH}(T, P) = V_{ref}(P) + \Delta V_{trap}(T, P) + \Delta V_b(T, P, \Delta V_{trap}) + \Delta V_{\mu}(T, P, \Delta V_b) \quad (1)$$

$$\sigma_{TH}(T, P) = \sigma_{ref}(P) + \Delta \sigma_{trap}(T, P) + \Delta \sigma_b(T, P) + \Delta \sigma_{\mu}(T, P) \quad (2)$$

Here, P denotes P1–P7, and $V_{ref}(P)$ and $\sigma_{ref}(P)$ are state-dependent references. The three contributions are physically coupled and jointly determine the descriptor evolution of the V_{TH} distribution during cross-temperature operation.

A. Statistical Broadening From Trap Coexistence

The trap term models the trap response during temperature transition [11], [12], [13], [14], [15], [16]. With $\Delta T = T - T_{ref}$, the asymmetric temperature accumulation is written as

$$S(T) = k_+ \max(\Delta T, 0) + k_- \max(-\Delta T, 0) \quad (3)$$

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where k_+ and k_- are the high- and low-temperature-side growth rates. The resulting trap-induced charge-loss term [11], [12], [14], [17], [18] is expressed as

$$L_Q(T) = -L_{\max} \left[(1 - m_{fs})(1 - \exp(-S(T))) + m_{fs}(1 - \exp(-\eta S(T))) \right] \quad (4)$$

where $L_{\max}$ is the maximum charge-loss amplitude, m_{fs} the fraction of the slowly responding trap component, and $\eta \in (0, 1)$ its reduced response rate. The mean contribution is $\Delta V_{trap}(T) = -L_Q(T)$, and the trap-induced sigma broadening is written as

$$\Delta \sigma_{trap}(T) = A_{trap} \rho(T) (1 - \rho(T)) [\Delta V_{trap}(T)]^2 \quad (5)$$

where A_{trap} is the scaling coefficient of the trap-related sigma term, and $\rho(T) = 1 - \exp(-\lambda_{TE} |T - T_{prg}|)$. The factor $\rho(T)(1 - \rho(T))$ is introduced to capture cell-to-cell variation.

B. Temperature-Modulated Barrier Response

The barrier term describes the temperature-dependent grain-boundary barrier response in the polycrystalline-silicon channel [2], [8], [19], where $x_q(T) = L_Q(T)/L_{\max}$ is the normalized charge-loss level. Since trap response modifies stored charge and local electrostatics, $x_q(T)$ couples trap-induced charge loss to the effective grain-boundary barrier. The barrier kernel is

$$F_B(T) = \exp \left(E_B e^{\alpha_{qb} x_q(T)} \left(\frac{1}{k_B T} - \frac{1}{k_B T_{ref}} \right) \right) - 1 \quad (6)$$

where E_B is the baseline barrier strength, α_{qb} controls exponential charge-loss modulation, and T_{ref} sets $F_B = 0$. The mean contribution is then written as

$$\Delta V_b(T) = K_B (1 + \beta_{qb} x_q(T)) F_B(T) \quad (7)$$

where K_B is the barrier amplitude coefficient and β_{qb} describes the linear charge-loss modulation of this amplitude. For the sigma branch, the barrier-induced broadening is written as

$$\Delta \sigma_b(T) = B_b [\Delta V_b(T)]^2 \quad (8)$$

where B_b is the scaling coefficient. The squared form represents second-moment scaling, linking cell-to-cell barrier-sensitivity variation to distribution broadening.

C. Temperature-Dependent Mobility Kernel

The mobility term describes temperature-dependent channel mobility [2], [7], [8], [19]. Since grain-boundary barriers reshape readout conduction, $F_B(T)$ is included to capture barrier-modulated mobility mapping as

$$\mu_*(T) = T^{-m} \exp \left(-\gamma_\phi \frac{F_B(T)}{T} \right) \quad (9)$$

where γ_ϕ is the coupling strength from $F_B(T)$ to the mobility term. The mean contribution is

$$\Delta V_\mu(T) = M_\mu \left[\frac{\mu_*(T_{ref})}{\mu_*(T)} - 1 \right] \quad (10)$$

where M_μ is the mobility amplitude coefficient. For the sigma branch, the mobility-related broadening is expressed as

$$\Delta \sigma_\mu(T) = C_\mu [\Delta V_\mu(T)]^2 \quad (11)$$

where C_μ is the corresponding scaling coefficient.

III. RESULTS AND DISCUSSION

A. Experimental Characterization and Descriptor Extraction

Commercial X3 9070 232-layer TLC charge-trap 3D NAND chips from the same fabrication lot were characterized under HPLR and LPHR trajectories. The vendor-specified operating window is 0°C–70°C, while –55°C–125°C was used as a controlled extended characterization window. Temperature compensation was disabled. During measurement, only the NAND package was locally temperature-controlled, while the external test platform, controller, communication link, and data-acquisition path remained ambient. In TLC operation, P0 denotes the erase state, while P1–P7 denote the programmed states. Each block was programmed to a single target state for state-isolated distribution extraction. Three chips with two dies each were evaluated, and all results are aggregated over the tested dies. For model extraction, global nonlinear robust least-squares fitting was performed on ΔV_{TH} and $\Delta \sigma$, with η , α_{qb} , β_{qb} , γ_ϕ , and λ_{TE} shared across program states.

B. Prediction Accuracy and Ablation Analysis

Fig. 1(a)–(h) assess restricted-window prediction. Under both HPLR and LPHR, the full model matches the held-out descriptors beyond the fitting window. To quantify prediction quality, an absolute relative error (ARE) is defined as

$$\text{ARE} = \sum_{i=1}^n \frac{|D_{\text{pred},i} - D_{\text{exp},i}|}{n |D_{\text{exp},i}|} \quad (12)$$

where D denotes the target descriptor, ΔV_{TH} or $\Delta \sigma$.

For the mean shift (ΔV_{TH}), the proposed model yields low ARE values of 0.065 under HPLR and 0.009 under LPHR, lower than all ablation cases. Removing the barrier term increases the HPLR ARE to 0.810, confirming barrier-dominated mean prediction, with trap and mobility terms playing secondary roles. This dominant role is physically reasonable because grain-boundary barrier modulation controls effective channel conduction. The larger relative ΔV_{TH} of P1 than P7 in Fig. 1(a) and Fig. 1(b) agrees with Fig. 2(b), where the higher P1 E_B indicates stronger barrier sensitivity from weaker electrostatic screening by stored charge. For the sigma variation ($\Delta \sigma$), the proposed model gives ARE values of 0.062 under HPLR and 0.054 under LPHR. Unlike ΔV_{TH} , $\Delta \sigma$ prediction is trap-dominated, as removing the trap term increases the HPLR ARE to 0.599, with the barrier and mobility terms being secondary. The opposite signs of $\Delta \sigma$ for P1 and P7 in Fig. 1(c) and Fig. 1(d) are consistent with this trap-dominated evolution, as cooling exposes trap-related cell-to-cell variation in P1 but suppresses thermally activated and field-assisted trap broadening in P7, leading to positive and negative $\Delta \sigma$, respectively. The non-monotonic $\Delta \sigma$ in Fig. 1(g) and Fig. 1(h) indicates a crossover from

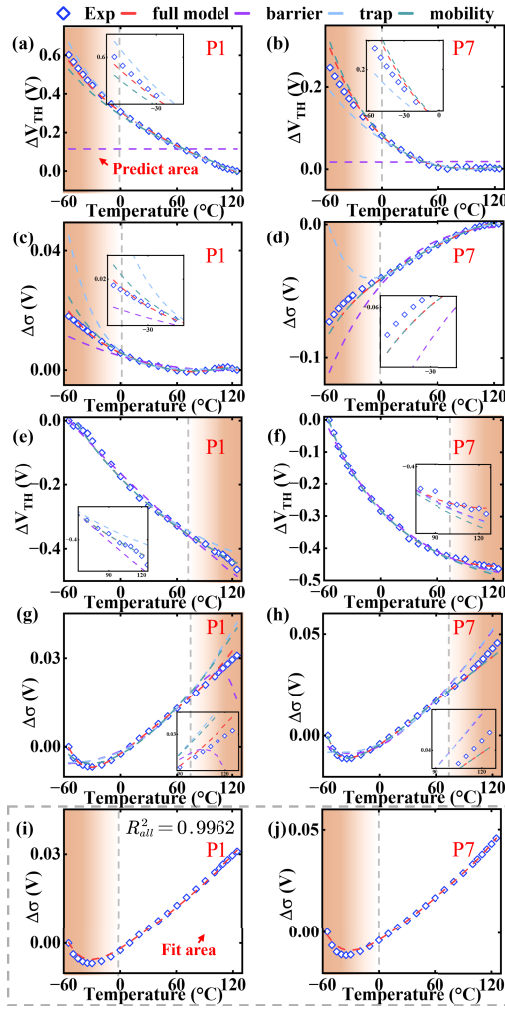


Fig. 1. Restricted-window fitting and prediction under HPLR and LPHR. (a)–(d) HPLR. (e)–(h) LPHR. (a),(e) P1 ΔV_{TH} . (b),(f) P7 ΔV_{TH} . (c),(g),(i) P1 $\Delta\sigma$. (d),(h),(j) P7 $\Delta\sigma$. Symbols/dashed lines denote experimental data/full model and ablations without the barrier, trap, or mobility term. Fitting window is 125°C to 0°C for HPLR and –55°C to 70°C for LPHR. In (i), (j) LPHR fitting uses the relatively linear 125°C to 0°C range, –55°C to 0°C is predicted.

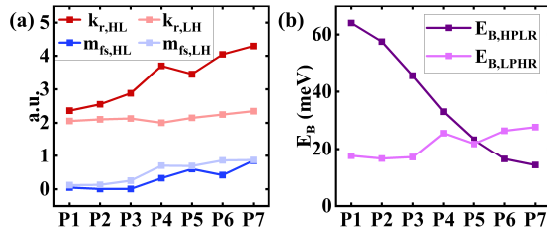


Fig. 2. Extracted parameters of proposed model under HPLR and LPHR. (a) State-dependent m_{fs} and k_r . (b) E_B versus program state.

mobility-related narrowing to barrier/trap-related broadening. Overall, the contributions of the three mechanisms can be ranked as barrier, trap, and mobility, with the barrier term dominating ΔV_{TH} prediction and the trap term dominating $\Delta\sigma$ prediction. Fig. 1(i) and (j) provide a stricter extrapolation test by excluding the nonlinear low-temperature region from fitting. The model still reproduces this region, confirming its effectiveness in strongly nonlinear extreme-temperature prediction.

Beyond prediction, Fig. 2 validates the physical expectations of the proposed model at the parameter level. Higher program

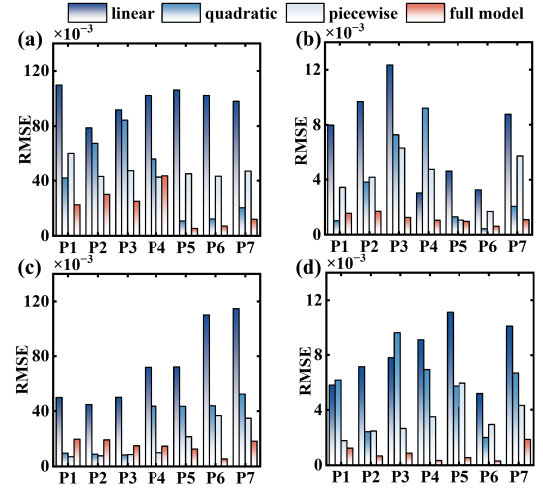


Fig. 3. Prediction-region RMSE comparison with empirical models. (a) HPLR ΔV_{TH} . (b) HPLR $\Delta\sigma$. (c) LPHR ΔV_{TH} . (d) LPHR $\Delta\sigma$. The fitting/prediction windows are 125°C to 0°C/0°C to –55°C for HPLR and –55°C to 70°C/70°C to 125°C for LPHR.

states are expected to contain a larger fraction of slowly responding traps, consistent with the extracted increase of m_{fs} with state. The effective barrier strength E_B is expected to be trajectory dependent rather than universally monotonic, and the opposite HPLR/LPHR trends support this expectation. As stored charge and path sensitivity increase toward higher states, stronger thermal-response asymmetry is expected, consistent with the increasing $k_r = k_+/k_-$. These trends support the trap–barrier–mobility picture.

The empirical baselines in Fig. 3 use linear $D(T) = aT + b$, quadratic $D(T) = aT^2 + bT + c$, and piecewise-linear fits with $D(T) = a_1T + b_1$ for $T < T_{tr}$ and $D(T) = a_2T + b_2$ for $T \geq T_{tr}$. Compared with these baselines under matched prediction windows, the proposed model yields lower RMSE for both ΔV_{TH} and $\Delta\sigma$ under HPLR and LPHR, reducing the average extreme-temperature prediction RMSE by 66.9%. This highlights the benefit of mechanism-constrained decomposition for held-out and nonlinear-region prediction. For controller use, the calibrated model may guide adaptive V_{read} lookup using temperature and thermal-history information.

IV. CONCLUSION

In this work, we proposed a mechanism-informed predictive model for the mean V_{TH} and sigma of 3D NAND threshold-voltage distributions across temperature. Full-window fitting showed the trap–barrier–mobility decomposition reproduces descriptor evolution over the measured temperature range. Restricted-window fitting, excluded-region evaluation, and nonlinear extrapolation verified prediction beyond the fitting window, including strongly nonlinear extreme-temperature regions. Under matched evaluation settings, the model achieved lower prediction error than empirical baselines for both mean V_{TH} and sigma under HPLR and LPHR. Although extracted parameters are specific to the tested 232-layer TLC charge-trap device, the framework may support recalibrated modeling of other charge-trap 3D NAND technologies. These results establish a compact distribution-level cross-temperature prediction framework for 3D NAND and guide read-reference adaptation, guard-band design, and product-specific cross-temperature calibration for improved wide-temperature robustness in commercial-grade 3D NAND.

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